Bogdan Dukic

Computer Vision | Computer Graphics | Machine Learning

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EDUCATION

École Polytechnique | Institut Polytechnique de Paris Sep 2026 to 2028 (expected)

MScT in Visual and Creative Artificial Intelligence

Merit scholarship: École Polytechnique Foundation Excellence Scholarship.

Current coursework: Machine Learning, Computer Animation, Digital Representations and Analysis of Shapes, Image Analysis and Computer Vision.

Upcoming coursework: Image Synthesis, Advanced Deep Learning, Multimodal Generative AI.

University of Novi Sad | Faculty of Technical Sciences Oct 2021 to Jul 2026

BSc in Computing and Control Engineering | **GPA: 9.55/10**

Intelligent Systems specialization (240 ECTS).

Scholarships: Dositeja (2025); Don Walker Scholarship, Magna International (2022 to 2025).

RESEARCH & SELECTED PROJECTS

Neuro-Symbolic Visual Scene Understanding Bachelor's thesis | 2025 to 2026

Python, PyTorch, YOLOv8, FastSAM, MiDaS, CLIP, T5 | Research summary

- Built an end-to-end visual question answering system combining object detection, segmentation and depth-based scene graphs with T5-generated symbolic reasoning programs.
- Adapted T5 via REINFORCE with execution-based rewards and KL regularization, improving end-to-end CLEVR-Humans accuracy from **32.7% to 45.6%** (7,202 questions) without target-domain program labels.
- Achieved **79.0% end-to-end accuracy** on CLEVR (10,000 questions); analyzed errors arising from visual perception and spatial relationships.

Federated Actor Learning Python, PyTorch, asyncio | 2026

- Implemented an asynchronous actor runtime with TCP messaging, supervision and SQLite persistence; built both centrally coordinated sample-weighted FedAvg and decentralized CRDT-based gossip training.
- Validated both training workflows on non-IID EuroSAT partitions using pretrained ResNet18 features, testing model exchange, aggregation and actor recovery.

Botanica Algorithmica | Procedural Plant Generation Python | Computer Graphics

- Created an interactive **2D procedural generation** tool using weighted L-system grammars, stack-based branching and controlled randomness to produce varied plant structures.
- Implemented growth visualization, editable branch and leaf parameters, plant presets and PNG export.

BunHub | Distributed Social Platform Java, Spring Boot, Angular

- Engineered a distributed social platform with real-time WebSocket messaging, RabbitMQ integrations and PostgreSQL-backed services.
- Implemented Bloom-filter username lookups, concurrency-safe social interactions using database locking, and scheduled background tasks.

PROFESSIONAL EXPERIENCE

AI Engineer | Comtrade System Integration Jan to Aug 2026 | Belgrade

- Developed the first working prototype of a no-code voice-agent platform in a four-person team, with visual scenario building, configurable guardrails and call analytics; integrated modular LLM, STT and TTS services.
- Built an automated document-processing pipeline for PDF, Word, TXT and Markdown, combining format-specific extraction, Docling and OCR with semantic chunking, hybrid retrieval and reranking for RAG applications.
- Implemented permission-aware retrieval using Microsoft Entra ID and automated SharePoint synchronization through Microsoft Graph API, webhooks and delta queries.
- Contributed to an agenda-aware Microsoft Teams meeting assistant; evaluated LLM, embedding and speech models for multilingual and Serbian-language applications.

Data Science Intern | Daze Solutions Limited Dec 2024 | Remote

- Built a real-estate analytics pipeline with PostgreSQL extraction, data cleaning, feature engineering, K-means clustering and Matplotlib visualizations using Python and Pandas.

TECHNICAL SKILLS & LANGUAGES

Programming: Python, C++, Java, SQL.
Machine Learning & Vision: PyTorch, Transformers, OpenCV, scikit-learn, NumPy, Pandas.
Graphics & tools: OpenGL, GLSL, Eigen, SFML, Git, Linux, Docker, Azure, PostgreSQL, FastAPI.
Languages: Serbian (native), English (C1; TOEFL iBT 100/120), French (A1, actively learning).